

Scope of Work — Grid Resilience Equity Strategy

Draft for Discussion

Parties

Consultant: Sandhya Persad **Client:** Addenda Capital **Date:** June 2026

Background

The Consultant has proposed a quantitative long/short equity strategy that trades U.S. utility stocks based on their measured sensitivity to electricity grid stress events. The strategy intends to construct a daily Grid Stress Index (GSI), the sensitivity of individual companies to that index to propose a trading strategy that rewards responsible grid investment.

The Client will provide access to Axioma risk model data. The Consultant will source all other required data from public and commercially available sources as described herein.

Scope of Work

Phase 1 — Data Integration & Signal Development

Estimated: 50–70 hours

The Consultant will:

1. **Establish and validate the data pipeline** — integrate public grid data (GridStatus, EIA API), equity price and fundamental data (yfinance), and short interest history (FINRA) into a reproducible research environment.
2. **Integrate augmenting data sources** — source forward power price data (CME DataMine electricity futures or Henry Hub gas futures as a validated free proxy) and wire into the signal layer. Source FINRA short interest history (free, twice-monthly publication) for the full universe.
3. **Develop enhanced signal components** — build and validate the following additional factor inputs alongside the existing stress beta:
 - Forward power price sensitivity (two-variable OLS: $GSI + \Delta \text{Forward Power Price}$)
 - Short interest momentum signal (3-month change in short interest as % of float)
 - Analyst revision momentum (3-month change in buy/sell ratio from recommendations data)
4. **Integrate Client-provided Axioma data** — align Axioma risk factor exposures to the universe and confirm data format compatibility.

Deliverable: working data pipeline, documented signal layer, initial factor scores with and without enhanced components.

Phase 2 — Backtest, Validation & Attribution

Estimated: 40–60 hours

The Consultant will:

1. **Run full rolling backtest** — extend the existing backtest framework to incorporate all active signal components; produce P&L, performance stats (Sharpe, maximum drawdown, etc), and monthly return series.
2. **Produce written findings report** — summarize performance results, signal construction, known limitations, and recommended configuration.

Deliverable: backtest results (code + output files), written findings report.

Phase 3 — Handoff & Documentation

Estimated: 20–30 hours

The Consultant will:

1. **Document all code** — provide inline documentation, function-level docstrings, and a technical reference covering each module, its inputs/outputs, and its role in the pipeline.
2. **Write operational runbook** — step-by-step instructions for running the pipeline, refreshing data, rebalancing, interpreting output, and handling common failure modes.
3. **Conduct handoff sessions** — participate in working sessions with the Client's designated technical contact to walk through the codebase, answer questions, and confirm the Client team can operate and extend the system independently.
4. **Deliver final codebase** — provide a clean, tagged release of the full repository in a format specified by the Client.

Deliverable: documented codebase, runbook, handoff session(s), final code delivery.

Compensation

Hourly Rate

\$150 per hour, billed against actual hours worked.

Monthly Hour Cap

Hours are capped at **30 hours per month**. Work exceeding 30 hours in any calendar month requires written pre-approval from the Client before the hours are incurred.

Invoicing & Timesheets

The Consultant will submit an itemized timesheet by the 5th business day of each month covering the prior month's work. Invoices are due net 30 days from submission. The Client may request a brief check-

in at any frequency to review hours against deliverables progress.

Data Cost Reimbursement

Out-of-pocket data costs (e.g., CME DataMine electricity futures subscription) will be billed as pass-through at actual cost, with receipts, and are **capped at \$600 total** across the full engagement. The Consultant will obtain written approval before incurring any single data expense exceeding \$100.

Data Responsibilities

The Client will provide data from Axioma risk model. All data sourced by the Consultant is drawn from public or commercially available sources. The Consultant will not use any material non-public information in the course of this work.

Intellectual Property

Upon receipt of full payment for each phase, the Consultant assigns to the Client all right, title, and interest in the work product produced under that phase, including source code, documentation, backtest output, and written findings. The Client will own all deliverables outright and may use, modify, and extend them without restriction.

The Consultant retains no rights to the deliverables and will not use Client-provided data (including Barra) for any purpose outside this engagement.

Scope Limitations

The following are explicitly outside the scope of this engagement:

- **Live trading or portfolio management.** The Consultant's role is limited to historical research and backtesting. All trading decisions are made exclusively by the Client.
 - **Investment advice.** Nothing in this engagement constitutes investment advice or a recommendation to buy or sell any security.
 - **Access to current or live positions.** The Consultant will work with historical and publicly available data only and will not have access to the Client's current portfolio, positions, or live trading systems.
 - **Performance guarantees.** Backtest results are historical simulations. Past simulated performance is not indicative of future results. No representation is made regarding live performance.
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Term

This engagement begins upon execution and continues until completion of Phase 3 or mutual written agreement to terminate, whichever comes first. Either party may terminate with 30 days written notice. Upon termination, the Client will owe fees for all hours worked and data costs incurred through the notice period, and the Consultant will deliver all work product completed to that point.

Signatures

Consultant

Signature: _____ Name: [Your Name] Date:

Client

Signature: _____ Name: _____ Title:

_____ Date: _____

This document is a draft scope of work and does not constitute a binding agreement until signed by both parties. Both parties should have counsel review prior to execution.